

Artificial Intelligence and Its Branches

Exploring Machine Learning, Deep Learning, NLP, Generative AI, and Agentic AI

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AI



ML



Deep Learning



NLP



Gen AI



Agentic AI

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is technology that enables computers to perform tasks typically requiring human intelligence — such as learning, reasoning, problem-solving, and decision-making.

Learning



Systems improve performance through experience and data analysis

Reasoning



AI draws conclusions and makes inferences from available information

Decision-Making



Systems evaluate options and select optimal actions to achieve goals

Perception



AI interprets sensory data like images, speech, and text

85%

of businesses use AI daily

50%

of emails are spam (blocked by AI)

\$1.8T

AI market to reach by 2030

Machine Learning

Machine Learning is a field of AI that involves developing algorithms and models that enable computers to learn and make predictions or decisions without being explicitly programmed.

Learning Process



Types of Machine Learning



Supervised Learning

Algorithms that learn from labeled data to make predictions or decisions



Unsupervised Learning

Algorithms that discover patterns in unlabeled data, such as clustering



Reinforcement Learning

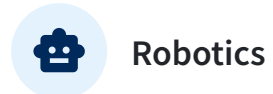
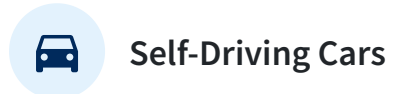
Algorithms that learn by interacting with an environment and receiving rewards or penalties

Deep Learning

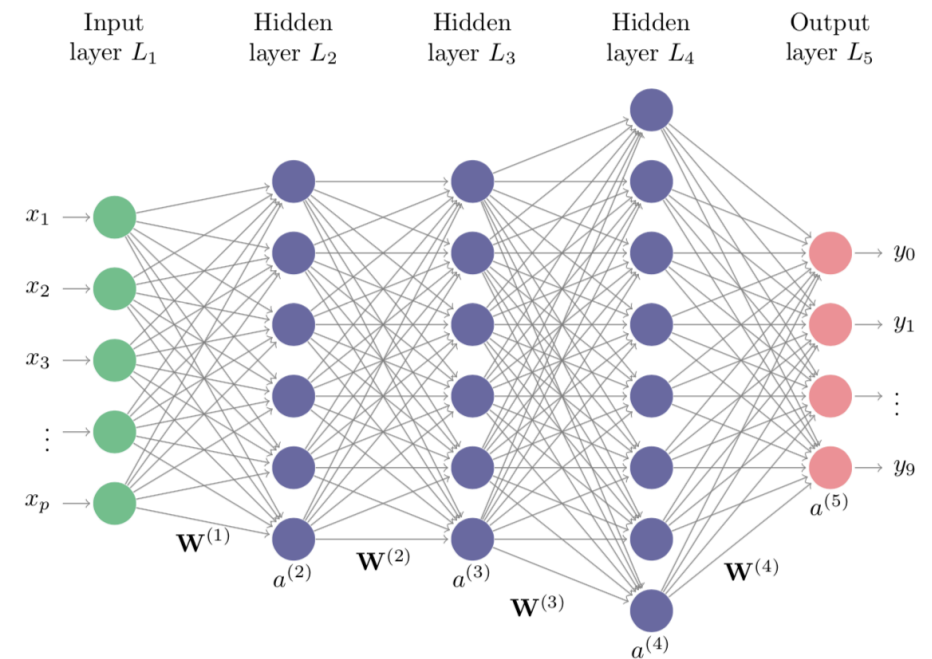
What is Deep Learning?

A **subset of machine learning** that uses multilayered neural networks inspired by the human brain structure

Key Applications



Neural Networks



Neural Network Structure: Input Layer, Hidden Layers, and Output Layer

Natural Language Processing (NLP)

🗣️ What is NLP?

Technology that allows computers to **interpret, manipulate, and comprehend human language** by combining computational linguistics with AI

💎 Key Components

NLU

Natural Language Understanding

NLG

Natural Language Generation

📊 Key Applications



Language Generation



Question Answering



Text Classification



Sentiment Analysis



Machine Translation

Natural language processing (NLP) tasks

NLP is the process through which AI is taught to **understand the rules and syntax of language**, programmed to **develop complex algorithms to represent those rules**, and then made to **use those algorithms to carry out specific tasks** like these.



Language generation



Answering questions



Text classification



Sentiment analysis



Machine translation

—zapier

NLP Tasks: Language Generation, Question Answering, Text Classification, Sentiment Analysis, and Machine Translation

Generative AI

✦ What is Generative AI?

AI that can **create new content and ideas** including conversations, stories, images, videos, and music

⚙️ How It Works

Uses deep-learning models trained on vast amounts of data to generate new, original content

1

Data Training

2

Pattern Learning

3

Content Generation



Key Applications: Content Generation, Music Creation, 3D Modeling, Video Creation, Game Development, Chatbots, Image Creation, Code Generation, Art Creation, Voice Generation

Agentic AI

⚙️ What is Agentic AI?

Autonomous AI systems that can act independently to achieve pre-determined goals with minimal human intervention

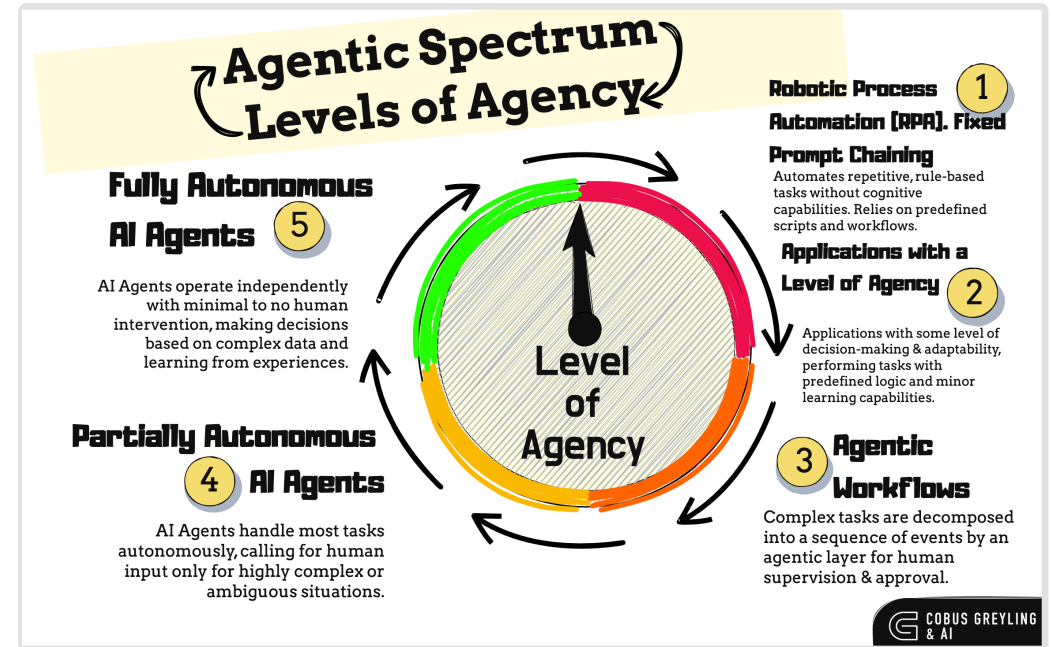
↔️ How It Differs

🔧 AI Agents

- ✓ Execute tasks
- ✓ Follow instructions
- ✓ Use preset tools
- ✓ Goal-oriented

🧠 Agentic AI

- ✓ Show autonomy
- ✓ Make choices
- ✓ Discover new methods
- ✓ Goal-driven



Spectrum of Agency: From RPA to Fully Autonomous AI Agents

Key Takeaways

1 AI is Already Here

AI is **integrated into daily life** — protecting inboxes, recommending entertainment, guiding travels, and saving lives in hospitals



2 Learning, Not Rules

Machine learning enables computers to **learn patterns from data** rather than following explicitly programmed rules



3 Diverse Branches

Each AI branch has **unique capabilities**:

- Deep Learning: Complex pattern recognition
- NLP: Human language understanding
- Generative AI: Content creation
- Agentic AI: Autonomous decision-making



4 Responsibility Matters

Responsible development of AI is essential for building trust and ensuring ethical use across all applications



Understanding AI is the first step toward using it effectively and ethically